

Phase 5 report – Generalization to other archetypes

Verdict: **PASS**

Subtask 5A – Simple series chain

An n -population feed-forward chain with equal inter-stage weight w and drive only on stage 0 has a recursive steady state $\rho_0 = f(I)$, $\rho_k = f(w \cdot \rho_{k-1})$. With no recurrence the Jacobian is triangular and all eigenvalues equal $-1/\tau$, so convergence to the steady state is exponential and free of oscillation; this is the natural population-level analogue of a feed-forward gain cascade.

The table below reports the maximum relative error between the analytical recursion and the numerical steady state over a weight sweep $w \in [0.2, 4.0]$ and chains of lengths $n \in \{2, 3, 5, 10\}$.

chain length	max rel. error over w -sweep
$n = 2$	$0.000e + 00$
$n = 3$	$1.431e - 13$
$n = 5$	$2.593e - 12$
$n = 10$	$9.450e - 11$

Acceptance (5 % relative error): **PASS** (overall max = $9.450e - 11$).

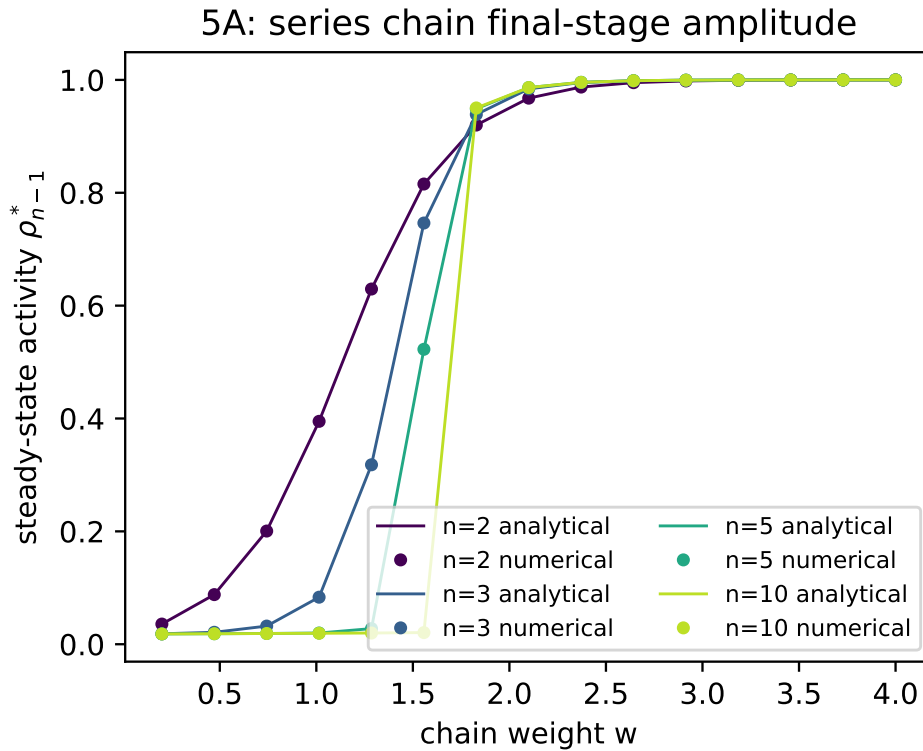


Figure 1: 5A. Final-stage activity $\rho_{\{n-1\}}^*$ versus chain weight w for chains of lengths 2, 3, 5, 10. Solid lines are the analytical recursive-sigmoid prediction; open dots are the numerical steady states. The curves overlap within line width.

Subtask 5B – Parallel composition

A driver population feeding n independently-weighted downstream populations produces a block-triangular Jacobian: the driver's 1×1 self-block is $-1/\tau$, the downstream $n \times n$ block is diagonal (no cross-coupling), and the only non-zero off-diagonal block is the driver $\rightarrow$ downstream coupling. Every eigenvalue therefore equals $-1/\tau$ exactly, independent of the gain vector $(w_i)_{i=1}^n$. The fixed point is closed-form: $\rho_0 = f(I)$, $\rho_k = f(w_k \rho_0)$.

n	FP L^∞ error	downstream off-diag mag.	downstream $\rightarrow$ driver coupling	eigenvalue error
$n = 2$	$0.00e + 00$	$0.00e + 00$	$0.00e + 00$	$0.00e + 00$

$n = 4$	$0.00e + 00$	$0.00e + 00$	$0.00e + 00$	$0.00e + 00$
$n = 8$	$0.00e + 00$	$0.00e + 00$	$0.00e + 00$	$0.00e + 00$

Acceptance (structure to machine precision, FP within 5 %): **PASS**.

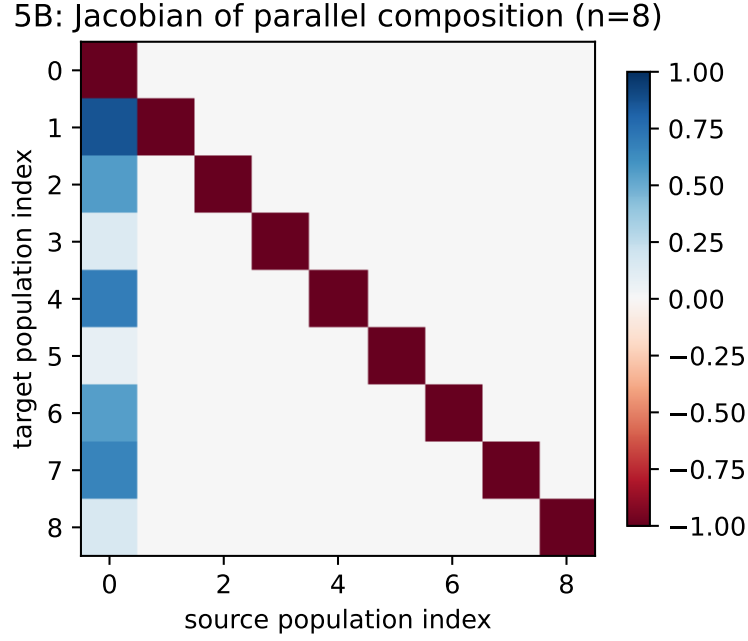


Figure 2: 5B. Jacobian of a parallel composition with $n = 8$ downstream populations. The block-triangular structure is visible: the driver's row (row 0) depends only on itself; each downstream row (rows 1–8) has a non-zero entry only on its own diagonal and on column 0.

Subtask 5C – Positive loop saddle-node bifurcation

Two mutually exciting populations with $w_{12} = w_{21} = w$ and zero drive admit a symmetric scalar reduction $\rho = f(w\rho)$. The saddle-node fold of the FP curve occurs when the slope condition $wk\rho(1 - \rho) = 1$ is met simultaneously with the FP equation. Eliminating ρ gives the analytical saddle-node weight(s) at $I = 0$, tabulated below against a numerical FP-count scan.

analytical w_{SN}	numerical transition	rel. error
1.6818	1.6756	0.371%
5.2778	5.2733	0.086%

Acceptance (5 % relative error): **PASS**.

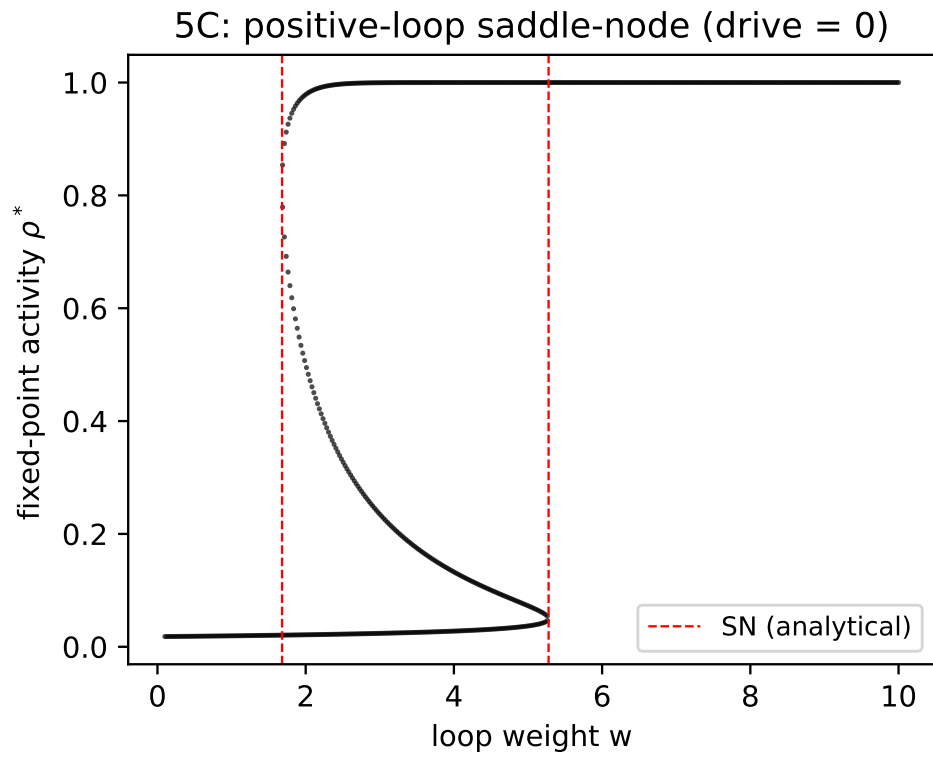


Figure 3: 5C. Fixed-point branches of the symmetric positive loop as functions of the loop weight w , at zero drive. A single low-active branch persists for small w ; above the analytical saddle-node weight (red dashed line) a high-active branch and a middle saddle appear.

Verdict

- 5A series chain: PASS
- 5B parallel composition: PASS
- 5C positive loop saddle-node: PASS

Overall: **PASS**.